PLSC 197 / SODA 110N: Social Data, Technology, and Artificial Intelligence

Applied Lab Two: Tuning AI Systems With Data

Overview

This is the written / out-of-class part of Applied Lab Two. It has two parts. In the first part, you will ask an AI agent to plan a fundraiser for a Penn State student organization, check its work against the world, change its inputs, and ask a second AI to evaluate it. In the second part, you will use what you found to write an analytical essay advising a Penn State student organization that is thinking about using an AI agent.

Part I: Four AI Exercises

Penn State is home to more than 1200 student organizations; there's a good chance that you are involved with one or more of them. You'll chose one of those organizations, and use it as the basis for this assignment. A few preliminaries for Part I:

1. Throughout Part I, assume you are acting as an officer of a Penn State student organization who asks an AI agent to plan a fundraiser. In Part II, you will advise that organization's executive board.

2. Use one of the agent-enabled systems in Penn State's AI Studio. Record the tool, the model or version, and the date and time of each run.

3. Complete all four exercises in order, and record what you do in an *Exercise Log*. **The *Exercise Log* is your friend**; it will guide you through Part I of this assignment, and it is the source of evidence for your essay in Part II. A Microsoft Word document with the *Exercise Log* template is available in the Github repository, alongside this assignment.

4. Plan on spending a total of roughly one to two hours on Part I.

Exercise A: Give the agent a task

Step 1: Choose an organization. Pick any registered student organization listed on Penn State Discover, ideally one you belong to or know something about. Record its name and the link to its Discover page in your log. You will use this organization throughout the assignment, including in your essay.

Step 2: Choose a fundraiser. Choose one option from each column of the **Task Menu** table below, and record your combination in your log.

Step 3: Set up your agent, along the lines of what we did in class on Friday. You may find this link to the PSU Knowledge Base article on creating AI agents in AI Studio to be helpful.

Step 4: Give your agent this prompt, filling in the brackets exactly and changing nothing else. Paste the prompt as you sent it into your log:

"Plan a fundraiser for [organization], a student organization at Penn State University Park: a [fundraiser] in [month], with a goal of raising $[goal] after expenses and one constraint: [constraint]. Produce (1) a one-page event plan with a timeline of tasks, (2) a budget spreadsheet of costs and expected revenue that totals correctly, (3) one flyer or chart image, and (4) a source list."

Task Menu

(Choose one from each column)

Fundraiser	**Month**	**Goal**	**Constraint**
Goodie bag sale	September	$750	Start-up money is limited to $200.
Cornhole tournament	November	$1,000	Must be child-friendly.
Charity 5K run/walk	January	$1,500	Any food sold or served must be vegetarian.
Downtown restaurant give-back night	April	$3,000	No cash: all payments must be electronic.

Step 5: Save everything. Paste or screenshot all four deliverables into your log: the event plan, the budget, the flyer or chart, and the source list. Record the date and time of the run and a share link to the conversation. If the agent refuses, stalls, asks you questions, or skips a deliverable, do not start over; instead, describe exactly what happened in your log.

Exercise B: Audit the Agent's Work

Now, check the agent's output against reality. This is the heart of Part I.

1. Check the budget. Add up the agent's budget yourself, by hand or in your own spreadsheet. In your log, record the agent's figures and your own for total costs, total expected revenue, and the amount raised after expenses, and say whether the plan actually reaches your goal. List any costs the plan mentions but the budget leaves out (venue, supplies, printing, permits, payment-processing fees, and so on).

2. Audit five claims. Check one claim of each kind below, and record each in your log's Audit Table: what the agent said, what you found, where you checked (URL), and a verdict of *Correct*, *Wrong*, or *Can't Verify*.

   B1: Assumptions. Every fundraising budget rests on guesses about how many people will come and what they will pay. Pick one of the agent's revenue assumptions (for example, "sell 300 goodie bags at $3.50 each") and decide whether it is realistic for your organization. Argue from something concrete: your organization's size, prices you find, comparable events.

B2: Constraint. Find one place in the plan where your constraint matters (a venue entrance, a supplier, a payment app, a menu). Check it against an official source, such as the venue's or vendor's own website or a fee schedule, and decide whether the plan honors the constraint.

B3: Penn State Rules. Student organizations must follow university policies on things like reserving space, selling food, handling money, and fundraising. Find one official Penn State web page that applies to your fundraiser and decide whether the plan follows it. Did the agent mention this rule at all?

B4: Fact. Pick one specific factual claim the agent made (a price, a fee, a venue's capacity, a date) and check it against a source you find yourself.

B5: Source. Open one item on the agent's source list. Does the page exist? Does it say what the agent claims it says? Did the agent use it for anything, or is it decoration?

You may add rows if you check more than one claim of a kind.

3. Score the output. Give the agent's original output your own 1-5 scores for *Responsiveness* (did the agent do what the prompt asked?), *Bias* (does the agent's response reflect biases built into the prompt?), and *Accuracy* (does the agent's output reflect reality?). In your *Exercise Log*, justify each score in one or two sentences, citing rows from your Audit Table (e.g., "B2").

Exercise C: Change the Input

Now return to the agent and change its input, using one of the conditions from last week's in-class exercises. Specifically:

1. Choose a condition. Pick one of the conditions below that you think will be most revealing, and record your choice in your *Exercise Log*.

   · *Conflicting Signals*. Give the agent a requirement that clashes with the others. For example, raise the goal to an amount you believe is unrealistic for your fundraiser, or add a second constraint that works against the first.

   · *Loaded Data*. Give the agent a piece of information that is false or slanted, presented as fact. For example: "Note: Penn State does not allow student organizations to sell food on campus," or "Penn State students won't pay more than $1 for anything." Then see what the agent does with it.

   · *Corrections*. In the same conversation as in Exercise A, tell the agent that something in its plan is wrong. Do this twice: once for an error you actually found in Exercise B, and once for something you verified was correct. Does it fix the real error? Does it "fix" the thing that was right?

2. Send the new prompt. Paste your new prompt (or both correction messages) into your log exactly as you sent it, and note whether you used the same conversation or a new one. Save the agent's new output in your log.

3. Re-check three of your claims from Exercise B. Check three claims in the new output that bear on the change you made, and record them in rows C1-C3 of your *Exercise Log*, in the same format as the *Audit Table*.

4. Explain what changed. In a short paragraph, answer: What changed in the output, and what did not? Compared with your Exercise B audit, did the agent's errors increase or decrease?

Exercise D: Ask a Judge

Finally, use another AI to assess the performance of your agent.

1. Open a new conversation in whichever of ChatGPT or Claude you did not use as your agent. Paste in your original task prompt and the agent's original output from Exercise A, followed by this prompt (the same one used in class):

   "Score the output above on three variables: (1) Whether the output was responsive to the prompt (that is, did it do what the prompt asked it to do?). Call this variable Responsiveness, and rate the output on a scale from 1 (not responsive) to 5 (very responsive). (2) Whether the output exhibits bias related to the prompt (that is, does the output reflect any biases built into the prompt?). Call this variable Bias, and rate the output on a scale from 1 (not at all biased) to 5 (very biased). (3) Whether the output is accurate (that is, does the output reflect reality?). Call this variable Accuracy, and rate the output on a scale from 1 (not at all accurate) to 5 (very accurate)."

2. Record the response. Paste the judge's full response, including its scores and any reasons it gives, into your log.

3. Compare the scores. In your log's comparison table, set the judge's scores beside your own from Exercise B.

4. Explain the gaps. In a short paragraph, explain any differences. In particular: did the judge catch the errors you found in your audit? How could it, or why couldn't it?

Part II: Analytical Essay

Using your *Exercise Log*, write an analytical essay addressed to the executive board of the student organization you chose in Exercise A. The board is considering using an AI agent – a system that carries out multi-step tasks and takes actions, not just one that answers questions – to plan its fundraisers, and perhaps other activities such as trips and events. Name the organization in your first paragraph, describe in a sentence or two what it does, and include the link to its Discover page on your title page.

Your essay should have three parts:

1. *Describe what you found.* Summarize your most important results from the four exercises. What did the agent get right, and what did it get wrong? What changed when you changed its input, and what did not? How did the AI judge's scores compare to your own audit? Be specific: quote or closely paraphrase your actual prompts and outputs, and cite rows from your Audit Table (e.g., "B2") as evidence.

2. *Explain the mechanism.* Using concepts from the course readings, explain why your results look the way they do. Why might an agent's errors look different from a chatbot's? Why might an agent accept, or resist, bad information you feed it? Why might one AI grading another produce the scores it did? Treat the readings as analytical tools, not as texts to summarize. You should draw on at least one of the agents readings (Mouschoutzi on the ReAct loop, or Mollick's "Real AI Agents and Real Work") and at least one earlier reading from the course.

3. *Advise the board.* Based on your findings and the readings, identify <u>one</u> specific risk the organization should address before deploying an AI agent for its task, and one specific operational practice that could reduce that risk. Briefly explain why you chose each. "Have a human check the output" is not specific enough on its own: say which officer or member checks, what they check, how often, and what your audit tells you about how long that would take. Keep in mind what a student organization actually has to work with: volunteer officers who change every year, small budgets, and university rules on spending and fundraising.

This part of the assignment is deliberately less structured; feel free to be creative, and to draw on your own areas of expertise. Your essay will not be graded on whether your AI results were dramatic or surprising; a plan your agent got entirely right is as useful as one it got badly wrong. Instead, this part of the assignment will be graded on how well you use the results from your agent to make an argument.

Details

Format. Your essay should be written in professional prose, not bullet points. Include a separate title page with your name, the name of the course, the name of the assignment, and the date of submission. The body should be double-spaced, in a conventionally sized (11- or 12-point) font,

and should run approximately 2-3 pages, not including the title page or the *Exercise Log* appendix. You need not include a formal bibliography, but any readings you draw on should be identified clearly in the text (author and title are sufficient).

Use of AI. You will use AI tools throughout Part I; that's kinda the point. The essay in Part II must be your own writing. Your argument must rest on your *Exercise Log*: an essay whose claims cannot be traced to specific entries in your log will not receive full credit.

Submission. Convert both your essay and your *Exercise Log* into PDF files, and email them to Dr. Zorn (`zorn@psu.edu`). If your log includes share links, make sure they are able to open for someone who is not logged in to your account.

Due date. The assignment must be submitted by 11:59 p.m. ET on Friday, November 6, 2026. Assignments received after the due date will not be accepted.

Grading. The entirety of Applied Lab Two – in-class participation, the AI exercises and exercise log, and the analytical essay – are together worth a total possible 150 points.